

DA3452 - Speech and Language Technology

Module 2 Test

Chapters 3, 4, 5 and Chapter 11 Sections 11.1.1-11.1.2 through TF-IDF

Duration: 60 minutes | Maximum marks: 45

Name: _____

Roll No.: _____

Date: _____

Instructions

Section A contains 15 applied multiple-choice questions of 1 mark each. Section B contains 15 advanced objective questions of 2 marks each.

Each MCQ has exactly one best answer. For fill-in questions, equivalent exact forms or decimals correct to three places are accepted.

Use log base 2 for entropy and natural logarithms unless another base is explicitly shown.

For TF-IDF, use $tf=0$ when $count=0$ and $tf=1+\log_{10}(count)$ otherwise; $idf=\log_{10}(N/df)$; $tfidf=tf \times idf$.

SECTION A

Questions 1-15 | 1 mark each | 15 marks

- An unsmoothed bigram model is trained on:
 - $\langle s \rangle a b \langle s \rangle$
 - $\langle s \rangle a c \langle s \rangle$
 - $\langle s \rangle b a \langle s \rangle$
- What probability does it assign to the sentence $a b$, including $\langle s \rangle$?
 - A. $1/6$
 - B. $1/9$
 - C. $2/9$
 - D. $1/3$
- For a history h , $C(h) = 18$, $C(hw) = 2$, vocabulary size $V = 12$, and $k = 0.5$. What is $P_{\text{add-k}}(w|h)$?
 - A. $1/12$
 - B. $5/48$
 - C. $1/8$
 - D. $5/36$
- An interpolated trigram model uses weights (0.2, 0.3, 0.5) for unigram, bigram, and trigram terms. For candidate a the probabilities are (0.10, 0.40, 0.20); for candidate b they are (0.40, 0.10, 0.20). Which result is correct?
 - A. $a: 0.24$; $b: 0.21$, so choose a
 - B. $a: 0.21$; $b: 0.24$, so choose b
 - C. $a: 0.22$; $b: 0.22$, so they tie
 - D. $a: 0.21$; $b: 0.24$, so choose a
- A source emits three symbols with probabilities (1/2, 1/4, 1/4). If a model represents this distribution perfectly, which pair gives (entropy in bits, perplexity)?
 - A. (1, 2)
 - B. (3/2, 3)
 - C. (3/2, $2\sqrt{2}$)
 - D. (2, 4)
- For a history h , $C(h) = 10$. Of $V = 5$ possible continuations, exactly two were observed, with counts 4 and 6. Under add-one smoothing, what total probability is assigned to the three previously unseen continuations?
 - A. $1/15$
 - B. $1/5$
 - C. $1/3$
 - D. $3/5$

7. A classifier currently has logit $z = 1.0$. A feature with weight -0.7 changes from 0 to 2; everything else is unchanged. Given $\text{sigmoid}(-0.4) = 0.401$, what is the new output at threshold 0.5?
 - A. 0.599; predict class 1
 - B. 0.599; predict class 0
 - C. 0.401; predict class 0
 - D. 0.401; predict class 1
8. A researcher trains on a training set, evaluates 40 configurations on the test set, and reports the lowest test perplexity. Which diagnosis is most accurate?
 - A. The estimate is pessimistic because repeated evaluation penalizes the selected configuration.
 - B. The estimate is unreliable because perplexity cannot compare smoothed configurations.
 - C. The estimate is optimistic because test performance influenced configuration selection.
 - D. The estimate is unbiased because every configuration used the same evaluation set.
9. A mini-batch contains $(y, \hat{y})$ pairs $(1, 3/4)$, $(0, 1/4)$, and $(1, 1/2)$. What is the mean binary cross-entropy loss?
 - A. $-\ln(9/32)$
 - B. $-(1/3)\ln(9/32)$
 - C. $-(1/3)\ln(3/32)$
 - D. $-(1/3)\ln(5/4)$
10. Initially $w = (0,0)$ and $b = 0$. A mini-batch has $x_1 = (2,0)$, $y_1 = 1$ and $x_2 = (0,1)$, $y_2 = 0$. Using the mean binary cross-entropy gradient and learning rate 0.2, what is (w,b) after one update?
 - A. $w = (0.10, -0.05)$, $b = 0$
 - B. $w = (-0.10, 0.05)$, $b = 0$
 - C. $w = (0.20, -0.10)$, $b = 0$
 - D. $w = (0.10, -0.05)$, $b = 0.10$
11. A nonzero co-occurrence vector v is replaced by $3v$ while nonzero u is unchanged. What happens to $u \cdot v$ and $\cos(u,v)$?
 - A. The dot product is unchanged; cosine triples.
 - B. Both triple.
 - C. The dot product is divided by 3; cosine is unchanged.
 - D. The dot product triples; cosine is unchanged.
12. On a validation set with 100 gold positives, threshold 0.3 gives $TP=90$ and $FP=90$; threshold 0.6 gives $TP=78$ and $FP=18$. A false negative costs 5 units, a false positive costs 1 unit, and recall must be at least 0.75. Among feasible thresholds, minimize total cost. Which tuple gives (selected threshold, recall, total cost)?
 - A. (0.30, 0.90, 140)
 - B. (0.30, 0.78, 128)
 - C. (0.60, 0.90, 140)
 - D. (0.60, 0.78, 128)
13. For one example, $\hat{y} = (0.6, 0.3, 0.1)$ and class 3 is correct. Starting from $b = (0,0,0)$, what is b after one gradient-descent step with learning rate 0.1?
 - A. (0.06, 0.03, -0.09)
 - B. (-0.06, 0.09, -0.03)
 - C. (-0.06, -0.03, 0.09)
 - D. (0.06, -0.03, -0.09)
14. A test set has 40 positive and 60 negative examples. A classifier predicts 30 positives, of which 24 are correct. Which tuple gives (precision, recall, F1, accuracy)?
 - A. (0.80, 0.60, 24/35, 0.78)
 - B. (0.60, 0.80, 24/35, 0.78)
 - C. (0.80, 0.60, 0.70, 0.84)
 - D. (0.75, 0.60, 2/3, 0.78)
15. A query vector q has norm 5. Candidates A-D have $(q \cdot c, \|c\|)$ equal to A:(18,4), B:(21,5), C:(12,3), D:(25,6). Which has the highest cosine with q ?
 - A. Candidate A
 - B. Candidate B
 - C. Candidate C
 - D. Candidate D

SECTION B

Questions 16-30 | 2 marks each | 30 marks

16. An add-one-smoothed bigram model is trained on:
 $\langle s \rangle a b \langle s \rangle$
 $\langle s \rangle a a \langle s \rangle$
 The predicted-token vocabulary is $\{a, b, \langle s \rangle\}$, so $V=3$. What perplexity does it assign to a b when a, b, and $\langle s \rangle$ are the three scored tokens?
 A. cube root of 10
 B. $\sqrt{10}$
 C. 10
 D. cube root of 30
17. An interpolated trigram model uses weights (0.2, 0.3, 0.5). For three scored positions, the (unigram, bigram, trigram) probabilities are (0.25, 0.50, 0.60), (0.50, 0.20, 0.18), and (0.25, 0.10, 0.09). Write (sequence probability, perplexity): _____.
18. A source distribution over $\{x, y, z\}$ is $p = (1/2, 1/4, 1/4)$, while a model assigns $m = (1/4, 1/2, 1/4)$. Using base-2 logs, write $(H(p), H(p, m), 2^{H(p, m)})$: _____.
19. A 5-gram model has training perplexity 2 but infinite development perplexity because a development 5-gram is unseen. An engineer applies stupid backoff and obtains 12 by inserting its S-scores into the perplexity formula. Which pair gives (source of the original failure, interpretation of 12)?
 A. (unseen-event zero probability, genuine perplexity)
 B. (unseen-event zero probability, non-probabilistic score)
 C. (test-set contamination, genuine perplexity)
 D. (test-set contamination, non-probabilistic score)
20. A two-example mini-batch has $x_1 = (1, 0)$, gold class 1, $y_{\hat{1}} = (1/2, 1/4, 1/4)$; and $x_2 = (0, 2)$, gold class 3, $y_{\hat{2}} = (1/4, 1/4, 1/2)$. Rows of W are features and columns are classes. Using mean cross-entropy, write dL/dW : _____.
21. A binary logistic model has one weight w and bias b , initially 0. With learning rate 1 it processes $(x=1, y=1)$, then $(x=2, y=0)$, using gradients $(y_{\hat{1}} - y)x$ and $y_{\hat{1}} - y$. Given $\text{sigmoid}(1.5) = 0.818$ and $\text{sigmoid}(-0.318) = 0.421$, what follows after both updates for test $x=0$, using threshold 0.5?
 A. $w = -1.136$, $b = -0.318$; predict 0
 B. $w = -1.136$, $b = -0.318$; predict 1
 C. $w = -0.318$, $b = -1.136$; predict 0
 D. $w = 2.136$, $b = 1.318$; predict 1
22. Rows are predicted classes and columns are gold classes:

	A	B	C
pred A	30	5	0
pred B	10	15	5
pred C	0	10	25

 Write (macro-F1, micro-F1, class with lowest F1): _____.
23. For each SGNS case, $L = [-\ln \text{sigmoid}(p) + \sum_j \ln \text{sigmoid}(-n_j)]$. Case 1 has $p = \ln 3$, $n_1 = 0$, $n_2 = -\ln 3$. Case 2 has $p = 0$, $n_1 = \ln 3$, $n_2 = 0$. What is the total loss?
 A. $\ln(512/9)$
 B. $\ln(256/9)$
 C. $\ln(64/9)$
 D. $\ln(512/27)$
24. Each subgroup has 50 gold positives and 100 gold negatives. Entries are (TP, FN, FP, TN):
 threshold 0.30: $G_1 = (46, 4, 16, 84)$, $G_2 = (44, 6, 7, 93)$
 threshold 0.50: $G_1 = (43, 7, 10, 90)$, $G_2 = (39, 11, 5, 95)$
 threshold 0.70: $G_1 = (39, 11, 6, 94)$, $G_2 = (35, 15, 3, 97)$
 Require overall recall ≥ 0.80 and $FPR = FP/(FP + TN) \leq 0.10$ in each group; among feasible thresholds maximize precision. What should be selected?
 A. 0.30 with precision 90/113
 B. 0.50 with precision 82/97
 C. 0.70 with precision 74/83
 D. None is feasible
25. Skip-gram learns target embeddings W and context embeddings C . Let $W_t = (1, 1)$, $W_a = (2, 1)$, $W_b = (1, 0)$, and $C_t = (1, -1)$, $C_a = (-1, 1)$, $C_b = (0, 1)$. Among a and b, use cosine similarity to write (nearest to t using W only, nearest to t using $W+C$): _____.
26. For a fixed target w , the positive dot product is $w \cdot c_{\text{pos}} = 0$. The negative-sampling vocabulary contains only f and r, with counts 16 and 1, and $w \cdot c_f = \ln 3$ while $w \cdot c_r = -\ln 3$. One negative is sampled per

positive, with $q(u)$ proportional to $\text{count}(u)^\alpha$; $q(f)=16/17$ for $\alpha=1$ and $8/9$ for $\alpha=3/4$. For $L=-[\ln \text{sigmoid}(w \cdot c_{\text{pos}}) + \ln \text{sigmoid}(-w \cdot c_{\text{neg}})]$, which expected-loss comparison is correct?

- A. $\alpha=1$ gives the lower expected loss, by $8 \ln 3 / 153$.
 - B. $\alpha=3/4$ gives the lower expected loss, by $8 \ln 3 / 153$.
 - C. $\alpha=3/4$ gives the lower expected loss, by $8 \ln 4 / 153$.
 - D. The expected losses are equal because the positive dot product is unchanged.
27. Cosines are woman-engineer=0.30, woman-homemaker=0.70, man-engineer=0.65, and man-homemaker=0.25. Let $b(g)=\cos(g,\text{engineer})-\cos(g,\text{homemaker})$ and $\delta=b(\text{man})-b(\text{woman})$. If embedding-based rankings influence job recommendations, which tuple gives (δ , possible harm, issue with one bank vector for bank teller and river bank guide)?
- A. (0.40, representational, sense separation)
 - B. (0.80, allocational, sense conflation)
 - C. (-0.80, allocational, sense separation)
 - D. (0.40, allocational, sense conflation)
28. Two 1000-document corpora use terms (a,b). Their document frequencies are (10,100) in corpus I and (100,10) in corpus II. The same three focal texts have raw counts $X=(10,1)$, $Y=(1,10)$, and $Z=(10,0)$, giving TF vectors $X=(2,1)$, $Y=(1,2)$, and $Z=(2,0)$. Using the course TF-IDF convention and cosine similarity from Chapter 5, which tuple gives (which of Y and Z has larger cosine to X in corpus I, which has larger cosine to X in corpus II, $\cos(X,Y)$, which is the same in both corpora)?
- A. (Z, Y, $5/\sqrt{34}$)
 - B. (Y, Z, $5/\sqrt{34}$)
 - C. (Z, Z, $4/\sqrt{17}$)
 - D. (Y, Y, $1/\sqrt{2}$)
29. For one positive pair and one negative sample, $L = -[\ln \text{sigmoid}(c_{\text{pos}} \cdot w) + \ln \text{sigmoid}(-c_{\text{neg}} \cdot w)]$. If both $c_{\text{pos}} \cdot w = \ln 4$ and $c_{\text{neg}} \cdot w = -\ln 4$, what is L?
- A. $\ln(5/4)$
 - B. $\ln(25/4)$
 - C. $2 \ln 5$
 - D. $\ln(25/16)$
30. The negative-sampling vocabulary contains exactly two words, whose corpus counts are 16 and 1. SGNS samples noise words proportional to $\text{count}(w)^{0.75}$. What are their sampling probabilities, in the same order?
- A. (16/17, 1/17)
 - B. (4/5, 1/5)
 - C. (64/65, 1/65)
 - D. (8/9, 1/9)

END OF QUESTION PAPER